MATH 11 — ELEMENTARY STATISTICS

§1.2

Types of Data

Who we study, what we measure, and what kind of thing we get back

Today's Goal

After completing this section, you should be able to:

- Define and distinguish between populations and samples
- Distinguish between parameters and statistics
- Classify data as qualitative or quantitative
- Classify quantitative data as discrete or continuous

What You Will Be Graded On

LT.1 Types of Data

I understand the differences between various types of data (e.g., qualitative, quantitative, discrete, continuous, etc.). I can use this understanding to identify what type of data a given data set is.

Who Are We Talking About?

A report says 3 in 10 Fresno State students work full time. It was written after asking 200 students. Which group is the report about?

All of them — but only 200 were measured. Each group gets its own name.

Observations and Their Values

Definition 1.2.1

Data are collections of observations. More precisely, data refers to the values associated with these observations. *Data* is plural; *datum* is singular.

So “the data *are*,” not “the data is.”

The Whole Group

Definition 1.2.2

A **population** is the complete collection of all subjects being considered in a statistical study.

It is also the group your conclusions are meant to apply to.

A population need not be people. It can be batteries, fish, or storms.

Everyone, or Only Some?

Definition 1.2.3

A **census** is the collection of data from every member of the population.

Definition 1.2.4

A **sample** is a subcollection of members selected from a population.

How the Sample Gets Chosen

Definition 1.2.5

Sampling is a process by which a sample is selected from a population.

The sample is the *result*. Sampling is the *action*.

Specific sampling methods come in §1.4.

The Property You Record

Definition 1.2.6

In statistics, a **variable** is a characteristic or property that can be determined for each member of the population. It is usually represented by a capital letter such as X or Y.

A person's height, a person's ethnicity, or whether or not a car passes a smog check.

Which Number Can We Get?

The average height of all 25,000 students at a university, or the average height of 100 of them. Which can you actually compute?

Only the second. The first is what we *want*; the second is what we can *get*.

A Number From Each Group

Definition 1.2.7

A **statistic** is a numerical property of a sample.

Definition 1.2.8

A **parameter** is a numerical property of a population.

Telling the Two Numbers Apart

A company wants to know the average commute time of all of its employees.

Parameter. The average over every employee. It describes the whole population, and it is *estimated*.

Statistic. Measuring everyone is impractical, so we take 60 employees. Their average describes the sample, and it is *calculated*.

Finding the Parts of a Study

A campus research office surveys $2{,}600$ California community college students. Of those surveyed, $1{,}144$ report working more than 20 hours a week.

The office uses this to estimate that 44% of California community college students work more than 20 hours a week.

Name the **population**, **sample**, **variable**, **statistic**, and **parameter**.

The Five Parts

Recall: $1{,}144$ of $2{,}600$ students surveyed work more than 20 hours a week.

- **Population.** All California community college students.
- **Sample.** The $2{,}600$ students surveyed. Careful: $2{,}600$ is the sample *size*, not the sample.
- **Variable.** Whether or not the student works more than 20 hours a week.
- **Statistic.** $\frac{1144}{2600} = 0.44$, or 44% — *calculated*.
- **Parameter.** 44% of *all* such students — *estimated*.

Now You Try: Name the Parts

You Try 1.2.1. A clinic checks 350 of the dogs it treats and finds 91 overweight.

1. Population? **All dogs the clinic treats.**
2. Sample? **The 350 dogs checked.**
3. Variable? **Whether the dog is overweight.**
4. Statistic? $\frac{91}{350} = 26\%$**, calculated.**
5. Parameter? **The percentage overweight among all of them.**

Is That Number a Number?

Your student ID is written with digits. Does adding two student ID numbers tell you anything?

No. Digits alone do not make data numerical. What matters is whether the values *count* or *measure* something, or merely *name* it.

When the Numbers Mean Amounts

Definition 1.2.11

Quantitative data (also called **numerical data**) consists of numbers representing counts or measurements.

When reporting quantitative data, always include appropriate units.

When the Values Are Names

Definition 1.2.12

Qualitative data (also called **categorical data**) consists of names or labels that represent categories.

A label may still be written with digits. That does not make it a measurement.

Sorting Numbers From Labels

1. The number of text messages a student sends in a day. **Quantitative** — it is a count.
2. The brand of phone a student uses. **Qualitative** — it names a category.
3. The ZIP code of a student's home address. **Qualitative.** Be careful here. It is written with digits, but it labels a place. Adding two ZIP codes produces nothing useful.

Now You Try: Numbers or Names?

You Try 1.2.2. For each variable, decide whether the data it produces is quantitative or qualitative.

1. The price of a textbook, in dollars. **Quantitative.**
2. A patient's blood type. **Qualitative.**
3. A runner's bib number in a race. **Qualitative.**
4. The number of pets in a household. **Quantitative.**

The test: would arithmetic on the values mean anything?

Values You Can Count

Definition 1.2.14

Discrete data is quantitative data that can only take on a finite or countable number of values.

A rule of thumb: data that comes from *counting* is usually discrete.

Values You Can Measure

Definition 1.2.15

Continuous data is quantitative data that can take on any value within a range, including fractional values.

And data that comes from *measuring* is usually continuous.

Sorting Counts From Measurements

1. The number of eggs a hen lays in a month. **Discrete** — a count. You cannot have 4.3 eggs.
2. The weight of a bag of oranges. **Continuous** — it can be measured to any precision.
3. The temperature of a cup of coffee as it cools. **Continuous** — it passes through every value in between.

Now You Try: Counts or Measurements?

You Try 1.2.3. For each variable, decide whether the data it produces is discrete or continuous.

1. The number of songs on a playlist. **Discrete.**
2. The rain that falls in a storm, in inches. **Continuous.**
3. The age of a tree, recorded to the nearest year. **Continuous** — the recording is rounded, the age is not.
4. The number of students absent from class. **Discrete.**

Putting It Together